

1 Supplemental Material 1

1.1 Implementation Details

1.1.1 Proposed INR (CiNeVol)

The network architecture of the proposed INR is summarized in Table 1. The multi-resolution hash grid $\phi_{spatial}$ was configured to have $L = 16$ resolution levels with $F = 2$ -dimensional feature vectors, a hash table size of 2^{19} , a base resolution of $16 \times 16 \times 16$, and a target resolution of $2048 \times 2048 \times 2048$. This resulted in multiresolutional spatial coordinate encoding vectors of size 32, of which the bias field MLP Φ_B considered only the coarsest two hash grid levels (4 features). The spatiotemporal hash grids were configured to have $L = 32$ resolution levels with $F = 2$ -dimensional feature vectors, a hash table size of 2^{19} , a base resolution of $8 \times 8 \times 8$, and a target resolution of $32 \times 32 \times 16$. This resulted in encoding vectors of size 64 for each of the decomposed spatiotemporal grids. The slice indices k and frame indices t , were encoded with learnable embeddings of size 16. All MLPs (Φ_φ , Φ_ψ , Φ_V , Φ_B , Φ_C) consisted of a single hidden layer with 64 neurons followed by ReLU activations. The feature vector $\mathbf{f}$, outputted by Φ_V , had a dimension of 16. We used batches of 2^{15} pixels, 16 PSF samples per pixel, $\lambda_{TV} = 0.01$, $\lambda_{Bias} = 0.1$, $\lambda_{Corr} = 0.5$, $\lambda_{Jac}^\varphi = 0.01$, and $\lambda_{Jac}^\psi = 0.1$. As the real-time MRI exhibited stronger variations than the phantom datasets (e.g. challenging blood flow effects and motion phase noise/inaccuracies), we slightly adapted the loss contribution weights for these datasets using $\lambda_{Corr} = 0.2$ and $\lambda_{Jac}^\varphi = 0.0001$. To reduce optimization times, we computed stack intersection masks and included only training pixels with coverage by more than one stack. The models were trained for 500 iterations using an AdamW optimizer with a learning rate of $5e-3$ and a weight decay of $1e-2$ for the MLP network parameters. For inference, we used isotropic inference grids and 16 PSF samples. For the *Phantom (FB)* experiments, we considered the target motion phases as network input. For the breath-hold inference experiments, we queried predictions of a single cardiac cycle with 20 frames at breath-hold (sampling 20 equidistant cardiac phases φ within the range $[0, 1]$ and setting the respiration phases ψ to 0 for *Phantom (BH)* and 0.5 for *In-vivo*). The INR experiments were implemented using *Python* version 3.10.14 and *PyTorch* version 1.13.1. The decomposed hash encoding implementation relied on [1]. The specifications of the hardware are a 32-core AMD CPU @ 3.78 GHz and an Nvidia® A100 40 GB GPU running with CUDA® driver version 12.2.

1.1.2 Baseline INR

We considered a simplified INR, consisting solely of Φ_V , Φ_B , and Φ_C , as a baseline model (Table 2). Rather than performing explicit deformation field estimation, the encoded 5D inputs $\phi_{spatial}(x, y, z)$, $\phi_{ecg}(\varphi)$, and $\phi_{resp}(\psi)$ were directly passed to the intensity MLP Φ_V , resulting in a direct mapping from spatiotemporal space to the intensity reconstruction. Besides increasing the network depth of Φ_V to three hidden layers with 64 neurons, the remaining architecture and hyperparameter choices were configured as in Sec. 1.1.1 ($\lambda_{TV} = 0.01$, $\lambda_{Bias} = 0.1$, $\lambda_{Corr} = 0.5$).

Table 1 CiNeVol model architecture summary.

Layer	Specifications	Input dim.	Output dim.
Hash-grid encoding	xyz (16 levels, 2 feat., base res. 16)	3	32
	xyt1 (32 levels, 2 feat., base res. 8)	3	64
	xyt2 (32 levels, 2 feat., base res. 8)	3	64
	yzt1 (32 levels, 2 feat., base res. 8)	3	64
	yzt2 (32 levels, 2 feat., base res. 8)	3	64
	xzt1 (32 levels, 2 feat., base res. 8)	3	64
	xzt2 (32 levels, 2 feat., base res. 8)	3	64
Slice encoding (ϕ_{slice})	Embedding	1	16
Frame encoding (ϕ_{frame})	Embedding	1	16
Cardiac deform. MLP (Φ_{φ})	1 hidden layer, 64 neurons, ReLU	224	3
Resp. deform. MLP (Φ_{ψ})	1 hidden layer, 64 neurons, ReLU	224	3
Intensity MLP (Φ_V)	1 hidden layer, 64 neurons, ReLU	32	17 (21 for +seg)
Correction MLP (Φ_C)	1 hidden layer, 64 neurons, ReLU	48	1
Bias MLP (Φ_B)	1 hidden layer, 64 neurons, ReLU	20	1

Table 2 Baseline model architecture summary.

Layer	Specifications	Input dim.	Output dim.
Hash-grid encoding	xyz (16 levels, 2 feat., base res. 16)	3	32
	xyt1 (32 levels, 2 feat., base res. 8)	3	64
	xyt2 (32 levels, 2 feat., base res. 8)	3	64
	yzt1 (32 levels, 2 feat., base res. 8)	3	64
	yzt2 (32 levels, 2 feat., base res. 8)	3	64
	xzt1 (32 levels, 2 feat., base res. 8)	3	64
	xzt2 (32 levels, 2 feat., base res. 8)	3	64
Slice encoding (ϕ_{slice})	Embedding	1	16
Frame encoding (ϕ_{frame})	Embedding	1	16
Intensity MLP (Φ_V)	3 hidden layers, 64 neurons, ReLU	416	17
Correction MLP (Φ_C)	1 hidden layer, 64 neurons, ReLU	48	1
Bias MLP (Φ_B)	1 hidden layer, 64 neurons, ReLU	20	1

1.1.3 NeSVoR

We employed the default parameters of the NeSVoR framework (version 0.5.0) [2], while heuristically selecting the following hyperparameters to optimize reconstruction performance (Table 3). Each model was trained for 1000 iterations using a batch size of 2^{15} and 128 PSF samples. The bias field MLP received features from the two lowest resolution levels, and both slice- and z- feature vectors were set to a dimension of 16. The loss function weights for image regularization and bias field regularization were set to 0.01 and 0.1, respectively. As in the proposed INR setups, we considered stack intersection masks to only include training pixels with coverage by more than one stack.

1.1.4 SVRTK

We employed a multi-stage SVRTK reconstruction approach following [3]. In the static motion correction stage, we performed stack-to-stack registration between temporal

Table 3 NeSVoR model architecture summary.

Layer	Specifications	Input dim.	Output dim.
Hash-grid encoding	xyz (12 levels, 2 feat., base res. 15)	3	24
Slice encoding)	Encoding	1	16
Intensity MLP	1 hidden layers, 64 neurons, ReLU	24	17
Correction MLP	1 hidden layer, 64 neurons, ReLU	32	1
Bias MLP	1 hidden layer, 64 neurons, ReLU	20	1

average SAX, LAX, and VAX stacks. Opposed to Van Amerom *et al.*, who defined a region of interest based on manual masks of the fetal chest, we considered the intersection mask of input stacks and derived isotropic reconstructions with 1.48mm resolution in SAX orientation. The initial stack-to-stack registration was followed by seven global registration-reconstruction iterations, increasing the number of super-resolution reconstruction iterations from 10 to 20 in the final global iteration. Relying on SVRTK default configurations, we used 16 normalized mutual information bins, no motion smoothing, bias field smoothing with $\sigma = 20mm$, mask smoothing with $\sigma_{mask} = 4mm$, a temporal sinc PSF, image smoothing decreasing from $\lambda = 0.02$ in the first iteration to $\lambda = 0.01$ in the last iteration, and $\delta = 150$ for edge definitions. We configured the multi-resolution smoothing to be performed at three levels, used stack intensity matching (scaling average intensities to 700), and deactivated robust statistics in the static reconstruction stage. Assuming that cardiac motion phases were known or accurately estimated, we skipped the image-based heart rate estimation and inter-slice synchronization of [3], used the pre-extracted cardiac phases (scaled to range $[0, 2\pi]$), and computed slice-wise average RR intervals. In the dynamic motion correction stage, we reconstructed a 3D SAX volume with 17-20 cardiac phases and 1.48mm isotropic resolution. We provided cardiac segmentation masks for the definition of a reduced region of interest (ground-truth masks for the phantom data and 2D nnUNet model segmentation masks [4, 5] for the SAX stacks of the in-vivo data). Enlarged, smoothed ROIs were derived by first transforming all slice masks according to the slice transformations of stage 1, and then combining the transformed masks to a joint reconstruction mask. The resulting mask then underwent smoothing (radius 10mm and sigma=3.33mm), thresholding (0.0027), and morphological operations (4 erosion iterations and 9 closing iterations). Stack and slice transformations were initialized with the solutions of the static registration stage and then optimized for four global iterations, again increasing the number of super-resolution reconstruction iterations from 10 to 20 in the final global iteration. While we activated robust statistics for the phantom data and in-vivo subjects 2 and 3, this configuration caused strong artifacts for volunteer 1, for which we, therefore, present reconstruction results without robust statistics.

References

- [1] Xu, J., Fan, Z., Yang, J., Xie, J.: Grid4D: 4D decomposed hash encoding for high-fidelity dynamic gaussian splatting. <https://github.com/JiaweiXu8/Grid4D>. GitHub repository, accessed December 2024, commit a8992a9 (2024)

- [2] Xu, J., Moyer, D., Gagoski, B., Iglesias, J.E., Grant, P.E., Golland, P., Adalsteins-son, E.: NeSVoR: implicit neural representation for slice-to-volume reconstruction in MRI. <https://github.com/daviddmc/NeSVoR>. GitHub repository, accessed October 2025, branch v0.5.0 (2023)
- [3] Amerom, J.F., Lloyd, D.F., Deprez, M., Price, A.N., Malik, S.J., Pushparajah, K., Poppel, M.P., Rutherford, M.A., Razavi, R., Hajnal, J.V.: Fetal whole-heart 4D imaging using motion-corrected multi-planar real-time MRI. *Magn Reson Med* **82**(3), 1055–1072 (2019)
- [4] Full, P.M., Isensee, F., Jäger, P.F., Maier-Hein, K.: Pretrained nnU-Net model from the cMRI M&Ms challenge 2020. <https://zenodo.org/records/4288464>. Accessed January 2025, doi:10.5281/zenodo.4134720 (2020)
- [5] Full, P.M., Isensee, F., Jäger, P.F., Maier-Hein, K.: Studying robustness of semantic segmentation under domain shift in cardiac MRI. In: Puyol Anton, E., Pop, M., Sermesant, M., Campello, V., Lalande, A., Lekadir, K., Suinesiaputra, A., Camara, O., Young, A. (eds.) *Statistical Atlases and Computational Models of the Heart. M&Ms and EMIDEC Challenges*, pp. 238–249. Springer, Cham (2021)